NABDH: the grid signs its own data

A pulse-based trust layer for Tunisian low-voltage grid data

TSYP14 Technical Challenge, EnergyMind Tunisia (IEEE IES & PES). Phase 1 project proposal.

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Repository: https://github.com/RayenRomdhani/nabdh

1. Summary

Automated grid decisions are only as good as the data behind them. A frozen sensor, a drifting voltage transformer or a tampered meter can feed false numbers into billing, load forecasting or protection logic without anyone noticing. NABDH ("pulse" in Arabic) is a trust layer that checks every reading before it reaches those systems.

Its core observation comes from physics. The grid frequency wobbles by a few millihertz every second, identically for every device on the synchronous grid, and the voltage on a feeder has its own small wobble shared by every meter on that transformer. A real meter records both pulses automatically. Invented or replayed data cannot, however realistic its values look. NABDH combines this pulse check with two physics witnesses (Ohm's law along the feeder and the energy balance at the transformer) and sorts every meter-minute into one of eight verdicts, each with a confidence score, a plain-language explanation, and a route that says whether the AI may act alone or a human must decide.

On a simulated Tunisian feeder with 24 meters at 1 Hz, the Phase 1 prototype reaches 0.97 precision and 0.90 recall at a 0.18 % false-positive rate, and mistakes only 0.6 % of genuine grid events for bad data. A classic Isolation Forest on the same values reaches 0.08 precision and 0.05 recall and raises false alarms on 35 % of genuine grid events.

2. Scope

We monitor one **smart-metered residential neighbourhood**: an MV/LV transformer and the low-voltage feeder behind it, with 24 single-phase smart meters, eight of them with rooftop PV of the PROSOL Elec type. The transformer has its own meter, which serves as the trusted reference.

We chose this slice because it is where Tunisia's grid is growing fastest in number of data points (smart meters, rooftop PV, home IoT), where a wrong reading touches a customer directly (billing, disconnection), and where the operator has the least visibility today.

In scope: data trust for voltage, current, active power and frequency at 1 Hz; telling genuine grid events from data problems; telling data problems from real equipment faults at the customer connection; escalation and audit; a design for sharing between neighbourhoods.

Out of scope for Phase 1: three-phase unbalance, harmonics and power-quality indices, MV network faults, and forecasting. NABDH protects the data those systems use; it does not replace them.

3. Objectives

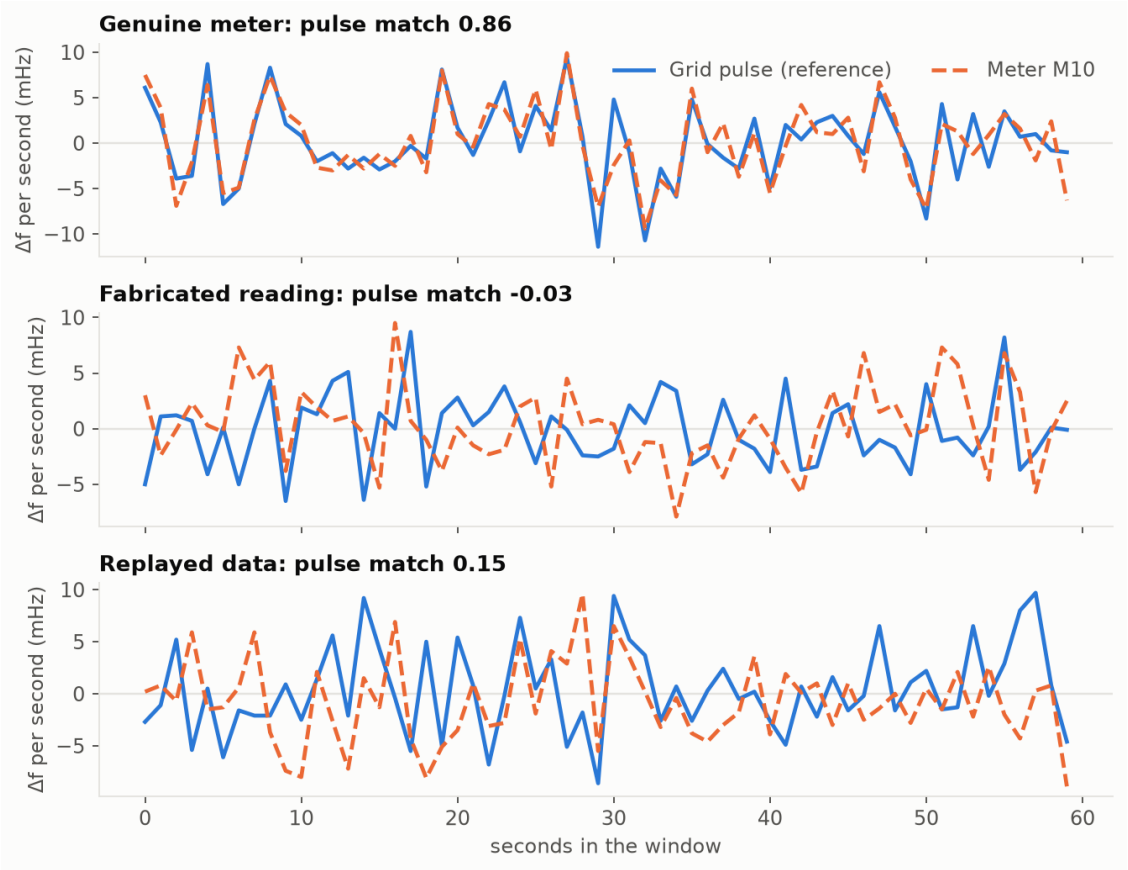
Goal	Measurable target	Phase 1 result
Detect problems	recall ≥ 0.85 on data-quality anomalies and equipment faults	0.90
Few false alarms	false-positive rate ≤ 1 % on normal windows	0.18 %
Real events stay real	≤ 5 % of genuine grid events flagged as data problems	0.6 %
Fast enough	median alarm latency ≤ 90 s, processing ≤ 1 s per minute of data	17 to 67 s, 6 ms
Explain everything	100 % of verdicts with confidence and plain-language reason	100 %
Humans own consequences	100 % of accusatory or asset actions routed to a human	100 %
Share safely	no raw consumption leaves a node	design (Phase 2 prototype)

4. The core idea: the grid's pulse

Frequency on an AC grid follows the balance between generation and load. When load rises slightly faster than generation, the rotating machines slow down and frequency drops by a few millihertz. When generation leads, it rises. Millions of small load changes make this a random signal, and because all synchronous machines are locked together, **every point of the interconnection sees the same random signal at the same instant**. Tunisia is synchronously linked with Algeria and Morocco and, through the Morocco-Spain interconnection, with the Continental European grid. Audio forensics has used this "electric network frequency" signature since the mid-2000s to prove when a recording was made. NABDH uses it to prove when a meter reading was made.

Voltage adds a second, local signature. Every meter on the same transformer hangs off the same secondary winding, so MV-side voltage changes and shared cable drops reach all of them together. After removing what each meter's own load explains (Ohm's law along the feeder), the remaining wobble is common to the whole feeder.

So a genuine reading carries two watermarks it cannot fake: the **grid pulse**, which proves the moment, and the **feeder pulse**, which proves the place. We compare second-to-second *changes*, never raw values. A meter drifting by 5 V still has the right pulse. A fabricated stream sitting at a perfect 231 V does not.



5. Data and anomaly-simulation plan

Fault-labelled grid data at one second does not exist publicly for Tunisia, so Phase 1 uses a seeded simulator (`nabdh/simulator.py`), as the challenge allows. It models grid frequency and MV voltage as mean-reverting random processes, households as base load plus cycling air conditioners or heaters, fridges and kettle/oven bursts, rooftop PV with a shared cloud factor, and the radial voltage drop along the feeder. Six one-hour episodes cover a summer evening, a heatwave, a Ramadan evening with the synchronised cooking surge before iftar, and a winter morning. Two clean episodes are used only for calibration.

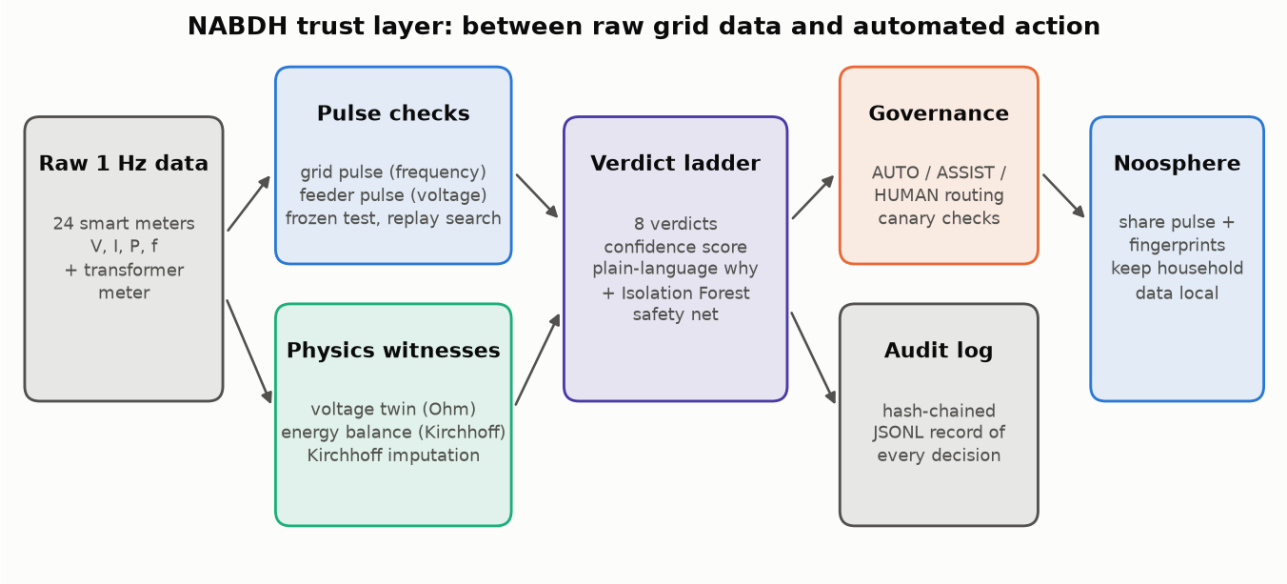
Genuine events are produced inside the physics: a 180 mHz frequency dip (generator trip elsewhere on the grid), a 10 % MV voltage sag for 20 s, and a loose customer connection (service resistance from 0.04 to 0.5 Ω with arcing). Data-quality problems are injected on the reported values only:

Injected	How
Stuck sensor	every channel repeats its value from the start of the fault
Drift (degraded sensor)	voltage offset ramps to +3 to +8 V
Bias (miscalibration or tampering)	current and power scaled by 0.4 to 0.7
Fabricated reading	synthetic voltage, power and a realistic 50 Hz wobble, made without grid access
Replay	the meter's own genuine data from 10 to 30 minutes earlier

The four test episodes hold 40 injections, 4 equipment faults and 8 grid events. Every one is listed in `data/ground_truth.csv`, and `data/README.md` documents the generation in full.

Phase 2 plan: replace the simulated frequency with a public 1 s Continental European frequency recording, shape household profiles on published STEG load curves, add a three-phase feeder, and add harder attacks (a forger who copies the public frequency, slow bias that never jumps).

6. Preliminary design for each challenge goal



6.1 Anomaly and bias detection

Every 60 s window (30 s hop) produces this evidence for each meter:

- **grid pulse match**: correlation of the meter's frequency changes with the transformer's;
- **feeder pulse match**: correlation of its voltage changes with its physics twin, i.e. the voltage rebuilt from the transformer voltage and every reported current;
- **frozen share**: fraction of seconds with no change at all;
- **local residual**: the meter's voltage minus its twin, compared with its neighbours on both sides, so a wrong current elsewhere (which shifts everyone downstream) cancels out;
- **bias onset**: when energy at the transformer suddenly stops being accounted for, the meter whose reading dropped by the same amount in the same second;
- **feeder context**: frequency excursion or transformer voltage sag.

A fixed rule ladder turns evidence into a verdict, in this order: STUCK, then the pulse test (REPLAY if the pulse matches an earlier moment, FABRICATED otherwise), then EQUIPMENT_FAULT and DRIFT from the residual, then BIAS from the energy balance, then GRID_EVENT. An Isolation Forest trained on clean windows is the safety net for anything the rules do not describe (UNEXPLAINED). Once a meter fails the pulse test, its current is rebuilt from Kirchhoff's current law (transformer current minus the trusted meters) so it cannot distort its neighbours' checks.

The distinction the challenge asks for falls out of the pulse. During a genuine event, the pulse stays coherent across every meter while the values move. During a data problem, one meter loses the pulse or breaks the physics while the rest of the feeder is calm.

6.2 Confidence scores and explainability

The evidence is the explanation. Each verdict's confidence is a logistic function of how far the deciding measurement sits beyond its threshold (for a replay, the match at the found delay). Thresholds come from clean data, so they are numbers an engineer can check. Each verdict comes with a sentence written for an operator, for example:

> *Meter M07: replayed data. The wobble is genuine but belongs to 1130 s ago (match 0.90 at that delay, 0.07 now). Old real readings are being re-sent as new ones.*

The operator interface shows the meter's pulse over the reference pulse for that minute, so a person can check the claim by eye. Every decision goes to a hash-chained log with the evidence, the explanation and the action. We use rule-backed scoring with physical thresholds by design: SHAP or LIME would explain a black box, and we do not need one.

6.3 Human-AI escalation and governance

The AI may always stop trusting data on its own, because that is reversible. Anything that acts on a customer or an asset needs a human.

Verdict	Action	Route
GRID_EVENT	accept data, alert operations	AUTO at confidence ≥ 0.80
STUCK	quarantine, maintenance ticket	AUTO ≥ 0.90, one-click ≥ 0.60
DRIFT	offset correction, recalibration	AUTO ≥ 0.95, one-click ≥ 0.70
BIAS	quarantine billing data, field inspection	always HUMAN
FABRICATED, REPLAY	quarantine, security incident	always HUMAN
EQUIPMENT_FAULT	crew to customer connection	always HUMAN

Consecutive windows merge into incidents: the test set produced 94 incidents, of which 29 were handled automatically, 1 went to one-click approval and 64 to the human queue.

Over-trust is measured with canaries: 5 % of the human queue are planted known-answer cases, the same idea as threat image projection in airport X-ray screening. Under-trust is measured as the share of correct verdicts the operator overrode. If over-trust rises, the interface hides the AI's recommendation until the operator has judged the plot. Details are in `docs/governance.md`.

6.4 Shared learning across the noosphere

Each neighbourhood, factory or microgrid runs its own node. The privacy line is simple: we share the heartbeat, not the household.

Shared	Kept local
grid pulse (frequency, identical everywhere, so not personal)	raw per-meter V, I, P (at 1 Hz it reveals presence and appliances)
grid event attestations	meter IDs, addresses, billing
anomaly fingerprints of confirmed incidents, with noise and k ≥ 5	local topology
calibration statistics	individual operator decisions

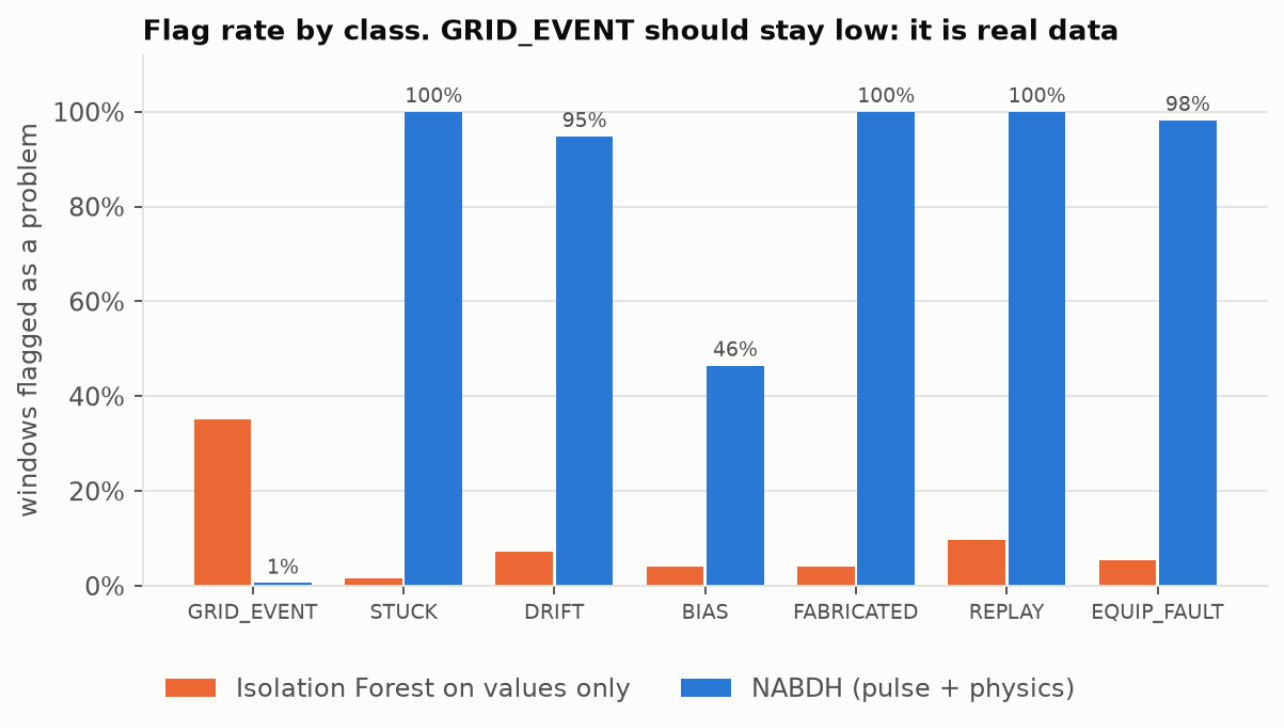
Shared pulses let nodes check each other's reference meters. Event attestations confirm that a dip is grid-wide. Matching fingerprints on several nodes reveal a coordinated campaign. Median-aggregated thresholds let a new node start without weeks of clean data while resisting a

poisoned peer. Raw consumption stays local because it is personal data under Tunisia's Organic Law 2004-63. The message formats and privacy guards are in `docs/noosphere.md`. Phase 1 runs a single node; the sharing layer is designed and will be prototyped in Phase 2.

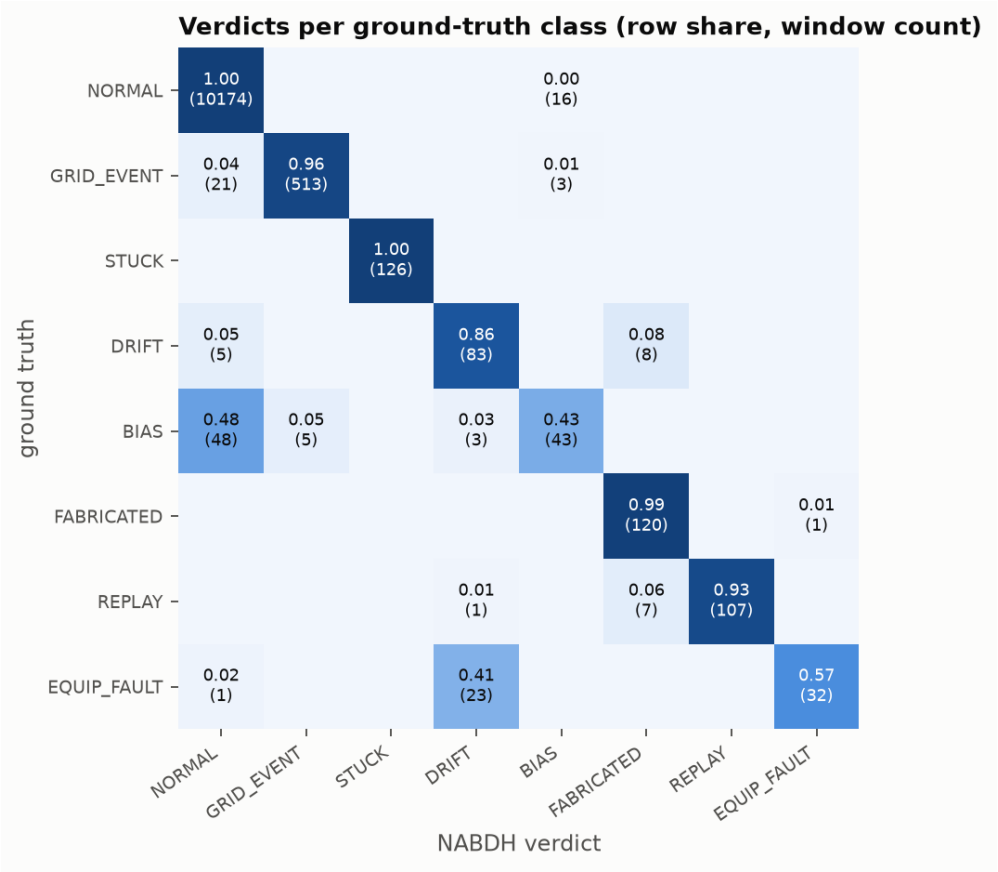
7. Phase 1 results

Test set: 4 episodes, 11,340 scored meter-windows.

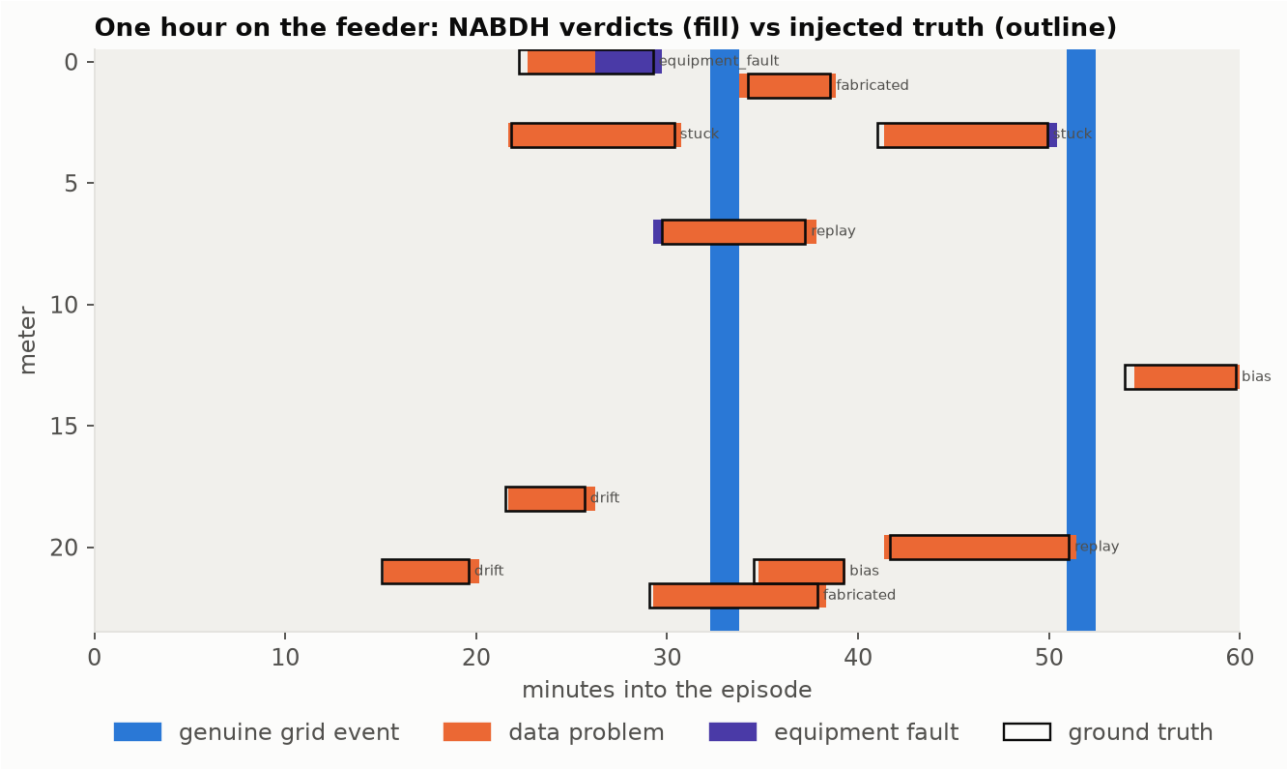
Metric	NABDH	Isolation Forest on values
Precision	0.967	0.084
Recall	0.904	0.052
F1	0.934	0.065
False-positive rate	0.18 %	3.2 %
Genuine grid events flagged as bad data	0.6 %	35 %



All 8 stuck, 8 drift, 8 fabricated and 8 replay injections and all 4 equipment faults were detected. Median latency from anomaly start to first alarm: 17 s for replay, 35 s for fabricated, 40 s for equipment faults, 50 s for stuck, 52 s for bias, 67 s for slow drift. One window of all 24 meters takes about 6 ms to process.



Two weaknesses are clear in the confusion matrix. Bias is caught in 46 % of windows (5 of 8 injections): a meter that draws little power, or exports PV, hides its bias inside the noise of the energy balance. Equipment faults are always flagged, but 41 % of their windows are labelled DRIFT, because when the house's current barely changes, a loose connection and a drifting sensor look alike. Both are first targets for Phase 2 (longer evidence windows and the meter's voltage response to its own load steps).



8. Business scope

These are planning estimates for a utility pilot, to be refined in Phase 2.

Deployment path. NABDH is software only. It needs 1 s voltage, current and frequency, which power-quality analysers, substation monitoring and advanced meter data concentrators already produce. A pilot starts with the transformer meter and the meters that can stream at 1 s, running on an edge PC at the substation or in a utility cloud.

Item	Estimate
Pilot integration (2 engineers, 4 months, 50 feeders)	60,000 to 90,000 TND one-off
Compute	under 1 TND per feeder per month (6 ms per minute of data)
Continuous training	automatic weekly recalibration on clean windows; 0.5 FTE data engineer per 1,000 feeders
Operator training and canary programme	2 days per control-room team
Licence model	per monitored feeder per month, about 15 to 25 TND

Value comes from three places: fewer wrong automated actions on bad data, earlier detection of meter tampering (non-technical losses), and earlier crew dispatch to dangerous loose connections. The pilot is meant to measure each of them on real feeders.

9. Plan for Phase 2

1. Operator web dashboard: incident queue, pulse plot, approve/override, canary dashboard.
2. Two- to five-node noosphere prototype with signed pulse and fingerprint exchange.
3. Real 1 s frequency data, STEG-shaped load profiles, three-phase feeder.

4. Better bias and fault separation; a forger who copies public frequency data.
5. Demo video and full report.

10. Risks and limits

- The method needs 1 s data. Many installed meters report every 15 minutes, so a first deployment covers feeders with power-quality or concentrator data.
- A forger with access to a real frequency stream could fake the grid pulse. The feeder pulse and the physics twin still catch this in our design, and it is a Phase 2 test case.
- Simulated data is cleaner than reality. Thresholds come from clean data and will need recalibration on real feeders.

References

1. C. Grigoras, "Digital audio recording analysis: the Electric Network Frequency (ENF) criterion", International Journal of Speech, Language and the Law, 2005.
2. F. T. Liu, K. M. Ting, Z.-H. Zhou, "Isolation Forest", IEEE ICDM, 2008.
3. TSYP14 EnergyMind Tunisia challenge specification, IEEE IES & PES, 2026.
4. Tunisian Organic Law 2004-63 on the protection of personal data.